

10. class, Electrodynamics and Optics

Page 869, Task 19.

An RLC circuit consists of a $150\text{-}\Omega$ resistor, a $21.0\text{-}\mu\text{F}$ capacitor, and a 460-mH inductor connected in series with a 120-V , 60.0-Hz power supply.

- (a) What is the phase angle between the current and the applied voltage?
- (b) Which reaches its maximum earlier, the current or the voltage?

Page 869, Task 25.

The LC circuit of a radar transmitter oscillates at 9.00 GHz .

- (a) What inductance is required for the circuit to resonate at this frequency if its capacitance is 2.00 pF ?
- (b) What is the inductive reactance of the circuit at this frequency?

Page 870, Task 28.

A $10.0\text{-}\Omega$ resistor, 10.0-mH inductor, and $100\text{-}\mu\text{F}$ capacitor are connected in series to a 50.0-V (rms) source having variable frequency. If the operating frequency is twice the resonance frequency, find the energy delivered to the circuit during one period.

Page 872, Task 50.

An $80.0\text{-}\Omega$ resistor and a 200-mH inductor are connected in *parallel* across a 100-V (rms), 60.0-Hz source.

- (a) What is the rms current in the resistor?
- (b) By what angle does the total current lead or lag behind the voltage?